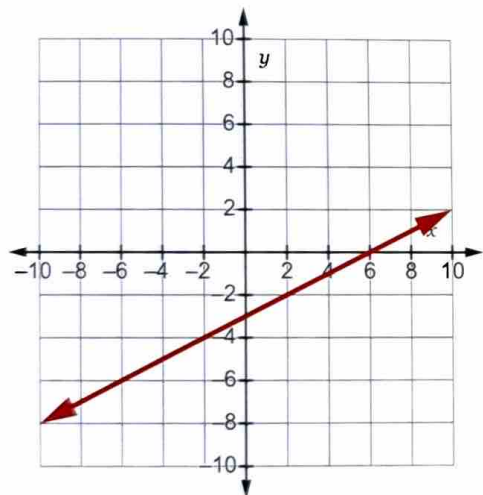


2026 Course 3 - Grade 8 Mathematics
Practice Test

The graph represents the linear equation $y = \frac{1}{2}x - 3$.



A second linear equation is represented by the data in the table.

x	y
-1	1
0	0
2	-2
3	-3

What is the solution to the system of equations?

(1, -1) ✖

(-2, -4)

(4, -4)

(2, -2) ✔

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ASSESSMENT
SUMMARY
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Practice Test

Consider this system of equations. Which equation represents the first equation written in slope-intercept form?

$$5x - 2y = 10$$

$$y = \frac{1}{4}x + 1$$

$$y = \frac{5}{2}x + 10$$

$$y = -5x - 5$$

$$y = -\frac{5}{2}x + 10$$



$$y = \frac{5}{2}x - 5$$

[RETURN TO
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SUMMARY](#)**33%**
SCORE

Four times a number added to 3 times a larger number is 31. Seven subtracted from the larger number is equal to twice the smaller number. Let x represent the smaller number and y represent the larger number. Which equations represent this situation?

$$y = -\frac{4}{3}x + 31$$
$$y = -2x + 7$$

$$y = -\frac{4}{3}x + \frac{31}{3}$$
$$y = 2x + 7$$

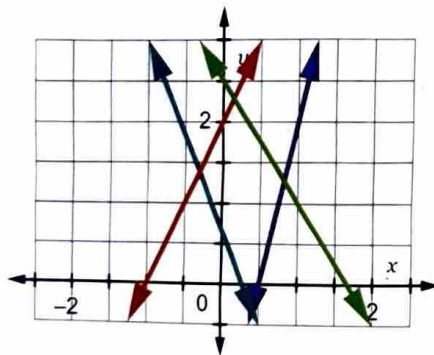
$$y = -\frac{4}{3}x + 31$$
$$y = 2x + 7$$

$$y = -\frac{4}{3}x + \frac{31}{3}$$
$$y = -2x + 7$$

RETURN TO ASSESSMENT SUMMARY
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SCORE

2026 Course 3 - Grade 8 Mathematics
Practice Test

The equations $2x - y = -2$, $3x + 2y = 5$, $4x - y = 2$, and $22x + 10y = 7$ are shown on the graph.



Which system of equations has a solution of approximately $(-0.3, 1.4)$?

$2x - y = -2$ and
 $3x + 2y = 5$



$4x - y = 2$ and
 $22x + 10y = 7$

$3x + 2y = 5$ and
 $4x - y = 2$

$2x - y = -2$ and
 $22x + 10y = 7$



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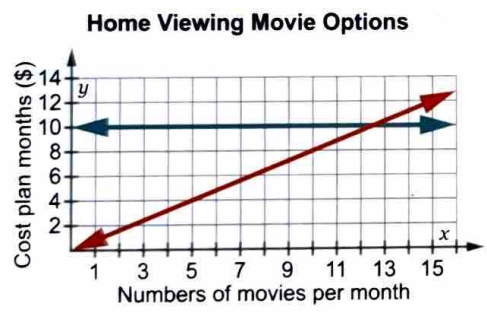


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Practice Test



Kedwin can watch movies at home in two ways. He can order unlimited online movies for \$10 a month. He could also go to the local movie rental store and rent 5 movies for \$4.00. The system representing this scenario is graphed as shown.

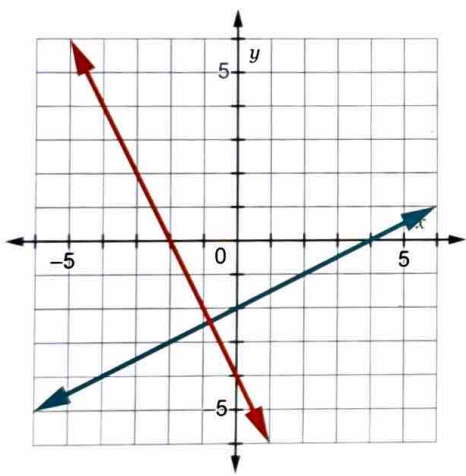


How many movies does Kedwin need to rent each month to make the online option a better deal?

- | | |
|------------------------------------|------------------------------------|
| 10.40 movies per month | 12.5 movies per month |
| 10 movies per month ✗ | 13 movies per month ✓ |

RETURN TO
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SUMMARY

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What system of equations is shown on the given graph?

$x - 2y = -4$ and $2x + y = 4$

$x - 2y = -2$ and $2x + y = -4$

$x - 2y = 2$ and $2x + y = 4$ ❌

$x - 2y = 4$ and $2x + y = -4$ ✅

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★ 33% SCORE

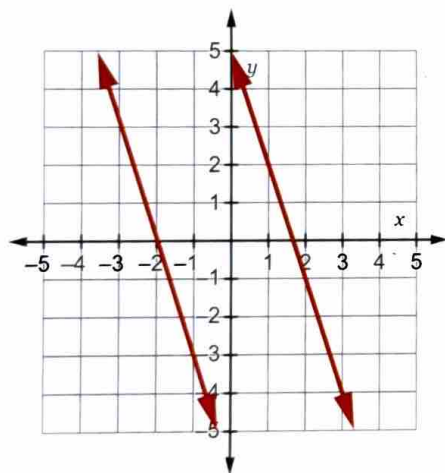




2026 Course 3 - Grade 8 Mathematics Practice Test



The system of linear equations $y = -3x + 5$ and $y = -3x - 6$ is graphed here.



How many solutions does the system of equations have?

3

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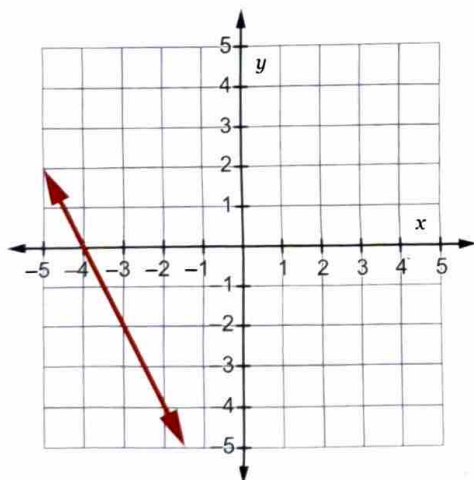


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Practice Test

The graphed line shown here is $y = -2x - 8$.



Which equation, when graphed with the given equation, will form a system that has infinitely many solutions?

$$y = -(2x + 8)$$



$$y = -2(x - 4)$$

$$y = -(-2x + 8)$$

$$y = -2(x - 8)$$

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Practice Test



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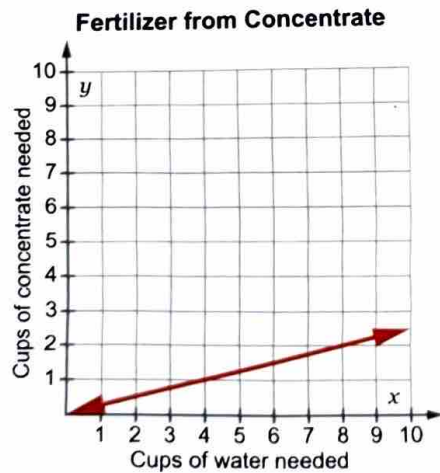
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Practice Test

Adam wants to mix 10 cups of fertilizer from concentrate. He needs to mix one part of fertilizer concentrate with 4 parts of water. He wrote the system of equations to represent the amount fertilizer concentrate and water needed and then graphed the first line.

$$y = \frac{1}{4}x$$

$$x + y = 10$$



Graph the second line. What is the solution to the system?

(4, 6)

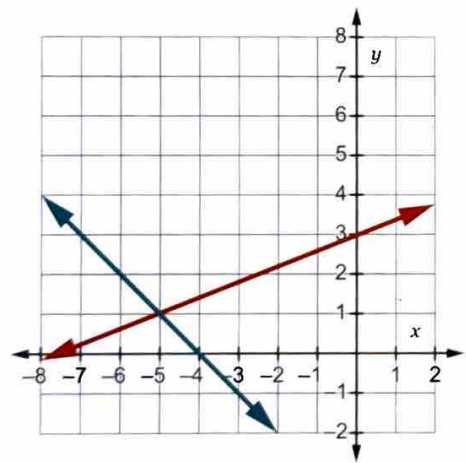
(0, 2.5)

(5, 5)

(8, 2) ✓

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What is the solution to the system of equations graphed here?

$(-4, 3)$

$(-5, 1)$ ✓

$(3, -4)$

$(1, -5)$ ✗

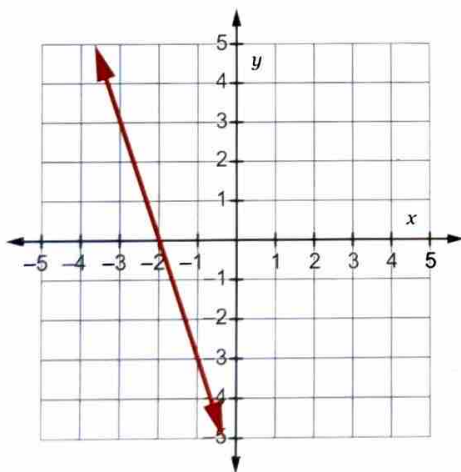
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Nita graphed the system of equations on the given graph.

$$y = -3(x + 2)$$

$$y = -3x - 6$$



Which explains why there is only one line?

Each has a slope of -3 and a y -intercept of -6 , so the system has no solution.



Each has a slope of -3 and a y -intercept of -6 , so the system has infinitely many solutions.



Each has a slope of -3 , but one has a y -intercept of 2 and the other has a y -intercept of -6 . So, the system has infinitely many solutions.

Each has a slope of -3 , but one has a y -intercept of 2 and the other has a y -intercept of -6 . So, the system has no solution.

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Practice Test

AR

Dimitri determined that the ordered pair $(2, -2)$ is a solution to the system of linear equations $7x + 9y = -4$ and $5x - 2y = 6$ as shown.

Dimitri's Work

$$7x + 9y = -4$$

$$7(2) + 9(-2) = -4$$

$$14 + (-18) = -4$$

$$-4 = -4$$

$$5x - 2y = 6$$

$$5(2) - 2(-2) = 6$$

$$10 - 4 = 6$$

$$6 = 6$$

What was Dimitri's mistake?

He checked the equation $7x + 9y = -4$ first when he should have checked $5x - 2y = 6$ first.

He made a mistake in his calculations when substituting the ordered pair into the equation $5x - 2y = 6$ and simplifying. ✓

He made a mistake in his calculations when substituting the ordered pair into the equation $7x + 9y = -4$ and simplifying. ✗

He mixed up the coordinates of the ordered pair when substituting it into the equations $7x + 9y = -4$ and $5x - 2y = 6$.

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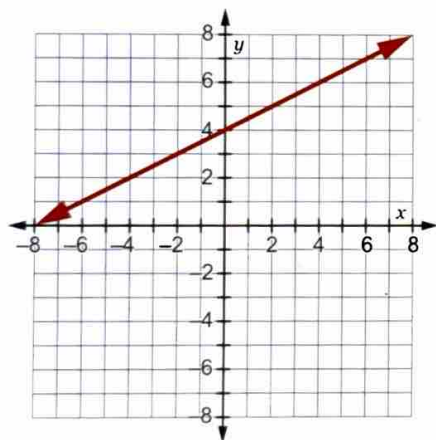


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2026 Course 3 - Grade 8 Mathematics

Representing Linear Systems Graphically



The equation $y = \frac{1}{2}x + 4$ is graphed here. Which equation would intersect this line at the point $(4, 6)$?

$$y = 4$$

$$y = 6x$$

$$y = 6$$



$$y = 4x$$

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

Test



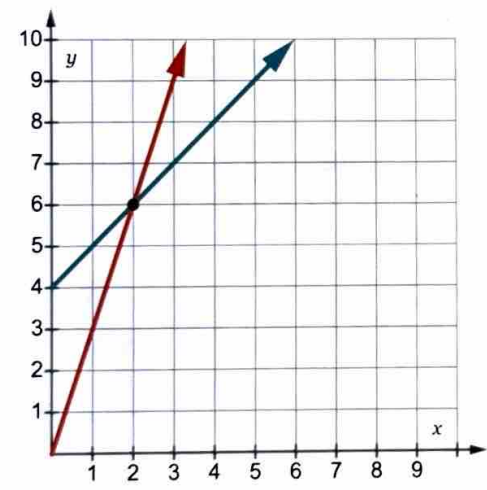
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2026 Course 3 - Grade 8 Mathematics
Representing Linear Systems Graphically



A number is equal to 3 times a smaller number. Also, the sum of the smaller number and 4 is the larger number. The situation is graphed on the coordinate plane, where x represents the smaller number and y represents the larger number.



Which two equations represent the situation?

$y = \frac{1}{3}x$ and $y = x + 4$

$y = 3x$ and $y = x + 4$ ✓

$y = 3x$ and $y = x - 4$

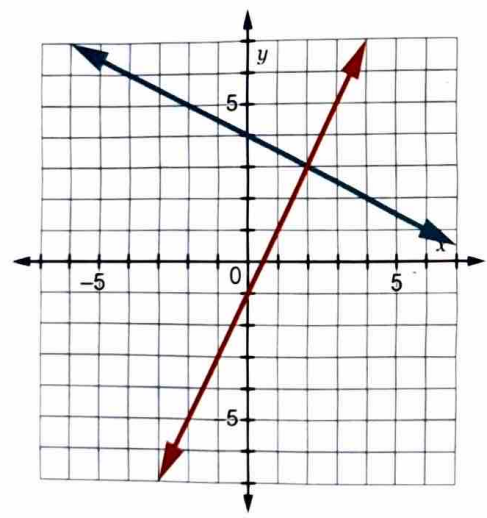
$y = \frac{1}{3}x$ and $y = x - 4$

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Representing Linear Systems Graphically

The system of equations $y = -\frac{1}{2}x + 4$ and $y = 2x - 1$ is shown on the graph.



According to the graph, what is the solution to this system of equations?

(4, -1)

(-1, 4)

(2, 3) ✓

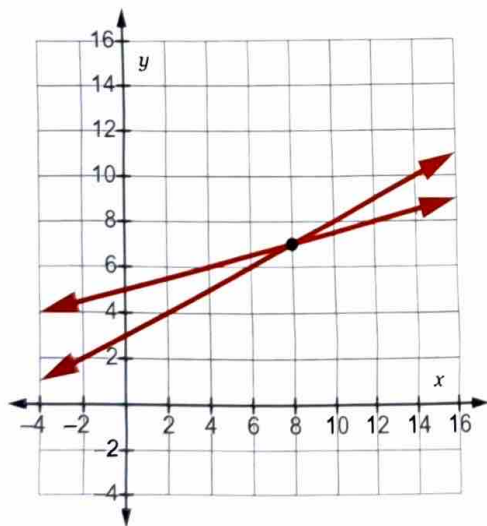
(3, 2)

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Representing Linear Systems Graphically

A number is equal to the sum of half a second number and 3. The first number is also equal to the sum of one-quarter of the second number and 5. The situation can be represented by using the given graph, where x represents the second number.



Which equations represent the situation?

$$y = 2x + 3 \text{ and } y = 4x + 5$$

$$y = 3x + 2 \text{ and } y = 5x + 4$$

$$y = \frac{1}{2}x + 3 \text{ and } y = \frac{1}{4}x + 5$$

$$y = 3x + \frac{1}{2} \text{ and } y = 5x + \frac{1}{4}$$

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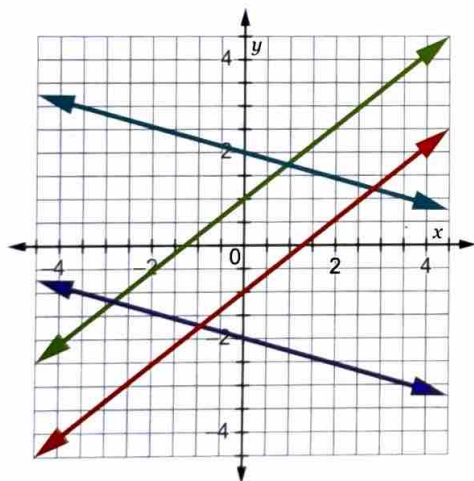
1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25



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Representing Linear Systems Graphically

The equations $4x - 5y = 5$, $3x + 10y = 20$, $4x - 5y = -5$, and $3x + 10y = -20$ are shown on the graph.



Which system of equations has a solution of approximately $(-2.7, -1.2)$?

$$4x - 5y = 5 \text{ and } 3x + 10y = 20$$

$$4x - 5y = -5 \text{ and } 3x + 10y = -20$$



$$4x - 5y = -5 \text{ and } 3x + 10y = 20$$

$$4x - 5y = 5 \text{ and } 3x + 10y = -20$$

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Representing Linear Systems Graphically

Jasmine and Madison are comparing babysitting rates. Jasmine makes \$2 for each child per hour. Madison makes a \$10 base rate plus \$4 per hour for any number of children. They defined x as the number of hours worked and y as the total money each earned. If they both watch 3 children, which statements are true? Choose four correct answers.

The money Madison earns can be expressed as $y = 10x + 4$.

The money Jasmine earns can be expressed as $y = 2x$.

The money Jasmine earns can be expressed as $y = 3(2x)$.

The independent variable is the number of hours worked.

The money Madison earns can be expressed as $y = 4x + 10$.

Madison will earn \$14 for each hour worked.

Jasmine will earn \$6 for each hour worked.

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Representing Linear Systems Graphically



AR

Anthony purchased 6 fruit bars and 3 chocolate-nut bars for a camping trip. He spent \$4.12 on the bars. Anthony wrote the following equation: $6x + 3y = 4.12$. What did he define as the variable x ?

number of fruit bars
purchased

cost of a fruit bar



number of chocolate-nut
bars purchased

cost of a chocolate-nut bar

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

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Which equation is written in slope-intercept form?

$3y = 8x - 5$

$y = 3x - 2$ ✓

$x = \frac{1}{2}y + 8$

$2x + 5y = 12$

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Representing Linear Systems Graphically



The equations in the given system are equivalent.

$$y = 2(x + 4)$$

$$y = 2x + 8$$

How many solutions does the system have?

no solution

an infinite number of
solutions



one unique solution

two solutions

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Representing Linear Systems Graphically

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Trang graphed the system of equations shown.

$$y = -2(x + 3)$$

$$y = -2x + 3$$

Trang says that the system of equations has no solution. Which explains whether or not he is correct?

Trang is not correct because the lines have the same slope but different y -intercepts.

Trang is correct because the lines have the same slope and the same y -intercept.

Trang is not correct because the lines have the same slope and the same y -intercept.

Trang is correct because the lines have the same slope but different y -intercepts. ✓



1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25



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Representing Linear Systems Graphically



Twice a number added to a smaller number is 5. The difference of 5 times the smaller number and the larger number is 3. Let x represent the smaller number and y represent the larger number. Which equations represent the situation?

$$2x + y = 5$$

$$x - 5y = 3$$

$$2y + x = 5$$

$$5x - y = 3$$



$$2x + y = 5$$

$$5y - x = 3$$

$$2y + x = 5$$

$$y - 5x = 3$$

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Representing Linear Systems Graphically

Aisha bought 15 apples and oranges for \$9.00. Each orange costs \$0.50, and each apple costs \$0.65. Let x represent the number of oranges and y represent the number of apples. Which system can be used to find the number of apples and oranges Aisha bought?

$$x = y$$

$$0.5x + 0.65y = 15.00$$

$$x + y = 15$$

$$0.5x + 0.65y = 9.00$$

$$x = y$$

$$0.5x + 0.65y = 9.00$$

$$x + y = 9$$

$$0.5x + 0.65y = 15.00$$

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

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Representing Linear Systems Graphically



Audra counted change collected for a class fundraiser. Audra wrote the equation $0.10x + 0.25y = 5.30$ to represent the situation. What did she define as the variable x ?

value of a quarter

value of a dime

number of quarters
collected

number of dimes collected ✓

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$$y = 3x + 8$$

no solution ✓

one unique solution

an infinite number of solutions

two solutions

an infinite number of solutions

two solutions



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Representing Linear Systems Graphically



Betty correctly determined that the ordered pair $(-3, 5)$ is a solution to the system of linear equations $6x + 5y = 7$ and $x + 4y = 17$. Based on this information, which statement is correct?

$(-3, 5)$ satisfies the equation $x + 4y = 17$ but not the equation $6x + 5y = 7$.

$(-3, 5)$ satisfies the equation $6x + 5y = 7$ but not the equation $x + 4y = 17$.

$(-3, 5)$ satisfies both the equation $6x + 5y = 7$ and the equation $x + 4y = 17$. ✓

$(-3, 5)$ satisfies neither the equation $6x + 5y = 7$ nor the equation $x + 4y = 17$.

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25



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Representing Linear Systems Graphically



Consider the system of equations shown here.

$$y = -5x + 1$$

$$y = -5x + 10$$

When graphed, the system consists of two lines that will never meet, no matter how far they are extended. Why are the lines parallel?

The linear equations have different slopes but the same y -intercept.

The linear equations have different slopes and y -intercepts.

The linear equations have the same slope but different y -intercepts. ✓

The linear equations have the same slope and y -intercept.

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Representing Linear Systems Graphically



Dominique needs to solve the system of equations by graphing. Which statement is true about the system?

$$2x - 3y = 12$$

$$y = -\frac{3}{5}x + 2$$

Neither equation is in slope-intercept form.

The second equation is $3x + 5y = 10$ when written in slope-intercept form.

The first equation is $y = \frac{2}{3}x - 4$ when written in slope-intercept form. ✓

Both equations are in slope-intercept form.

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2026 Course 3 - Grade 8 Mathematics

Representing Linear Systems Graphically

Which ordered pair is a solution to the system of linear equations $-4x + y = 8$ and $x - 5y = 17$?

$(-3, 4)$

$(-3, -4)$ ✓

$(-4, 3)$

$(-4, -3)$

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1 2 3 4 5 6 7 8 9 10 11 12 13 14 15 16 17 18 19 20 21 22 23 24 25

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Representing Linear Systems Graphically



Consider this system of equations. Which shows the second equation written in slope-intercept form?

$$y = 3x - 2$$

$$10\left(x + \frac{3}{5}\right) = 2y$$

$$y = 5x + 3$$



$$y = \frac{1}{2}x + 6$$

$$y = \frac{1}{5}x + \frac{3}{25}$$

$$y = 5x + \frac{3}{10}$$

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Representing Linear Systems Graphically



AR

Consider the system of linear equations $2x + 3y = 8$ and $3x + y = -2$. Which statement is correct?

The point $(1, 2)$ is a solution to the system of equations because it satisfies the equation $2x + 3y = 8$.

The point $(1, 2)$ is not a solution to the system of equations because it does not satisfy the equation $3x + y = -2$. ✓

The point $(1, 2)$ is not a solution to the system of equations because it satisfies neither equation.

The point $(1, 2)$ is a solution to the system of equations because it satisfies both equations.

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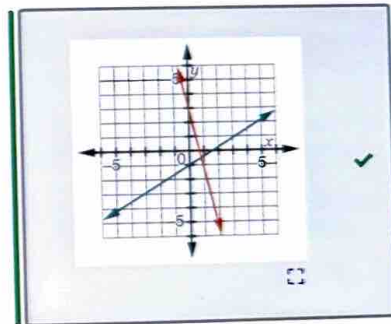
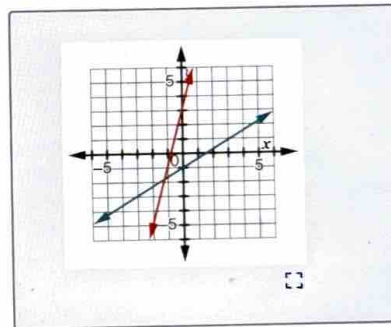
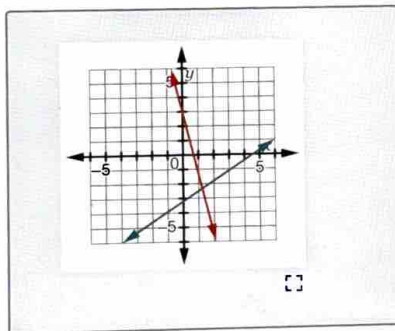
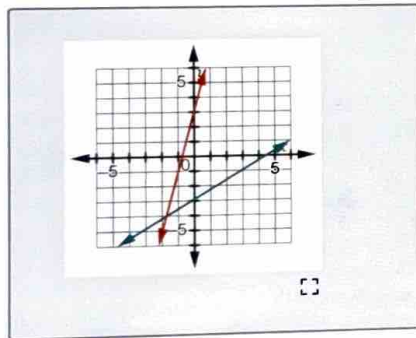
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Representing Linear Systems Graphically

Which graph shows the system of equations $4x + y = 3$ and $2x - 3y = 3$?



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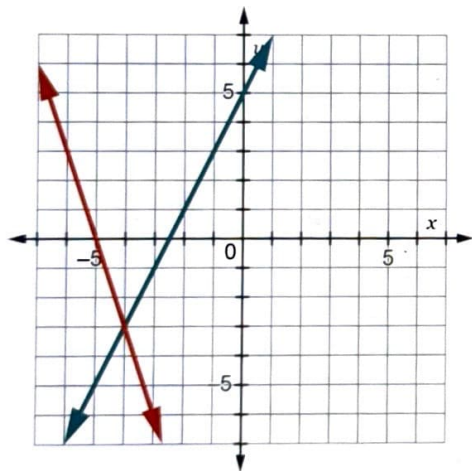
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Representing Linear Systems Graphically

The system of equations $y = 2x + 5$ and $y = -3x - 15$ is shown on the graph.



According to the graph, what is the solution to this system of equations?

$(-3, -4)$

$(-4, -3)$ ✓

$(5, -5)$

$(-5, 5)$

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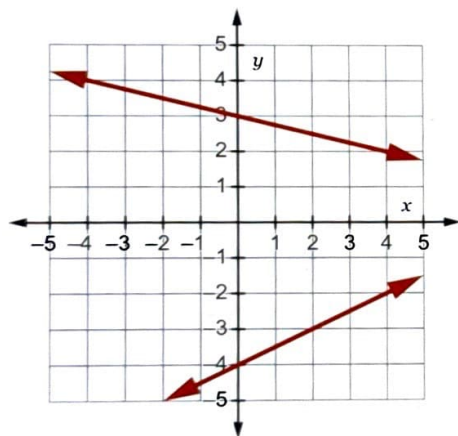
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Representing Linear Systems Graphically



Which statement is correct about the system of linear equations graphed here?

The system of equations has one solution because the lines will eventually intersect.



The system of equations has one solution because the lines will never intersect.

The system of equations does not have one solution because the lines will never intersect.

The system of equations does not have one solution because the lines will eventually intersect.

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Submitted

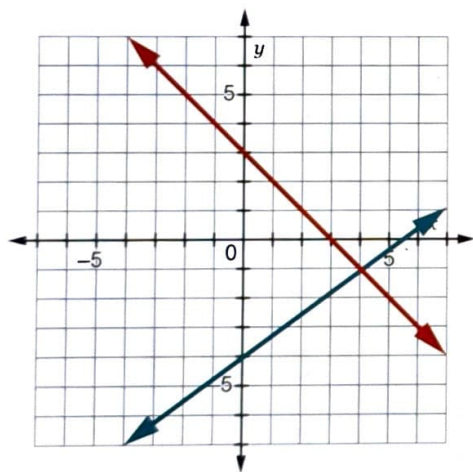


2026 Course 3 - Grade 8 Mathematics

Representing Linear Systems Graphically



The system of equations $y = \frac{3}{4}x - 4$ and $y = -x + 3$ is shown on the graph.



According to the graph, what is the solution to this system of equations?

$(-1, 4)$

$(4, -1)$ ✓

$(1, 4)$

$(4, 1)$

RETURN TO
ASSESSMENT
SUMMARY

★
96%
SCORE



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